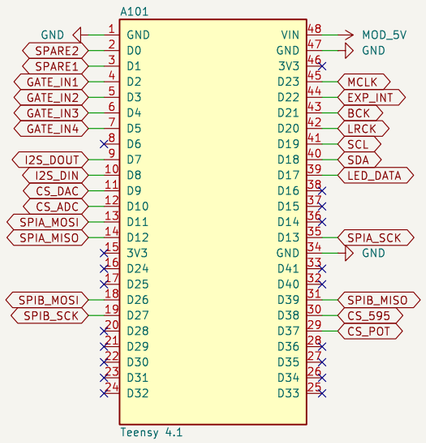
ANALOGUE BOARD REV A FOR TEENSY 4.1 — COMPUTE MODULE AND PIN MAP

Every signal into the module is 3.3 V.
The module's own 3.3 V output is not used: the board makes its own +3V3A and +3V3D.
Teensy 4.1 (not 5 V tolerant): cut the VIN-VUSB pads before fitting; VIN takes MOD_5V.
Every pin has its hole, so standard full-length headers fit; the ones this board does not use are left open.



MIDI: bridge SPARE1–MIDI_TX and SPARE2–MIDI_RX (adjacent holes)
with a wire or shunt. Left open, SPARE1/2 are plain GPIO.
On a Teensy they are Serial1 (TX1, RX1).

PIN MAP

Net	Function	Teensy 4.1	Why this pin
MCLK	I2S master clock to SCKI1/2	D23	MCLK1
BCK	I2S bit clock	D21	BCLK1
LRCK	I2S word clock	D20	LRCLK1
I2S_DOUT	I2S data to codec DIN	D7	OUT1A
I2S_DIN	I2S data from codec DOUT	D8	IN1
SPIA_SCK	Converter bus clock	D13	SCK
SPIA_MOSI	Converter bus data out	D11	MOSI
SPIA_MISO	Converter bus data in (ADS8688)	D12	MISO
CS_ADC	ADS8688 chip select	D10	CS
CS_DAC	DAC8568 SYNC	D9	
SPIB_SCK	Panel bus clock	D27	SCK1
SPIB_MOSI	Panel bus data out	D26	MOSI1
SPIB_MISO	Panel bus data in (MCP3208)	D39	MISO1 (alt pin)
CS_595	74HCT595 latch (RCLK)	D38	
CS_POT	MCP3208 chip select	D37	
SDA	I2C data	D18	SDA (Wire)
SCL	I2C clock	D19	SCL (Wire)
EXP_INT	TCA9555 interrupt, active low	D22	
GATE_IN1	Gate input 1	D2	
GATE_IN2	Gate input 2	D3	
GATE_IN3	Gate input 3	D4	
GATE_IN4	Gate input 4	D5	
LED_DATA	WS2812 data (buffered to 5 V)	D17	TX4 (WS2812Serial)
SPARE1	Spare GPIO 1, or MIDI out (link hole)	D1	TX1
SPARE2	Spare GPIO 2, or MIDI in (link hole)	D0	RX1
MOD_5V	Module 5 V in	VIN	Schottky D203

SHEETS

Power
Power entry, buck, linear rails, test points
File: power.kicad_sch

Codec & audio
PCM3060, two audio ins, two audio outs
File: codec_audio.kicad_sch

CV jacks & ADC
ADS8688, DG412 jack switches, CV holes
File: cv_io.kicad_sch

CV DAC
DAC8568 and the eight ±5 V output stages
File: cv_dac.kicad_sch

Pots & gate outs
MCP3208 pots, 74HCT595 gate outputs
File: pots_gates.kicad_sch

Expander, I2C & LEDs
TCA9555 buttons, EEPROM, Qwiic, WS2812
File: expander_i2c.kicad_sch

Gate ins
LM393 gate inputs
File: gate_in.kicad_sch

Edge strips
n8synth edge strips, fiducials
File: edge_strips.kicad_sch

TRS MIDI
MIDI in (H11L1) and out (5 V buffered), Type A
File: midi.kicad_sch

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Analogue board for Teensy 4.1. Spec: docs/analogue_board_spec.md
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Sheet: /
File: analogue_board_teensy.kicad_sch

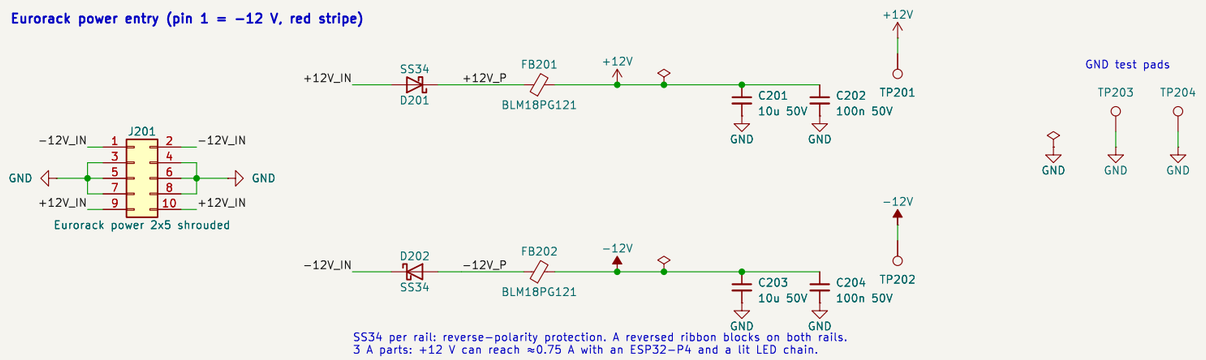
Title: Compute module and pin map

Size: A3	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 1/10

POWER

±12 V enter through a 2×5 shrouded header (the bus +5 V is not used). Each rail: series Schottky, ferrite, 10 µF + 100 nF. +5VD comes from a forced-PWM buck (no light-load pulse skipping in the audio band); +5VA from a linear regulator on its own; each 3.3 V rail from its own LDO. A 0 Ω link after every regulator lets a faulty rail be isolated and its current measured.

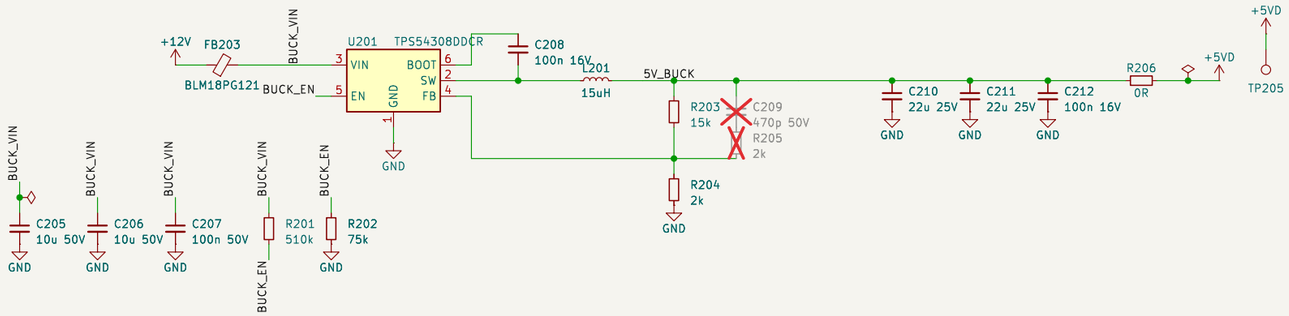
Eurorack power entry (pin 1 = −12 V, red stripe)



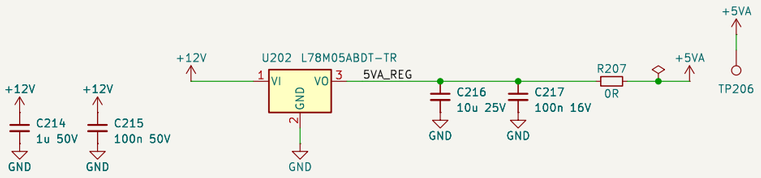
SS34 per rail: reverse-polarity protection. A reversed ribbon blocks on both rails. 3 A parts: +12 V can reach ≈0.75 A with an ESP32-P4 and a lit LED chain.

+5VD: TPS54308 forced-PWM buck, 350 kHz (TPS54202 is a pin- and value-compatible fallback)

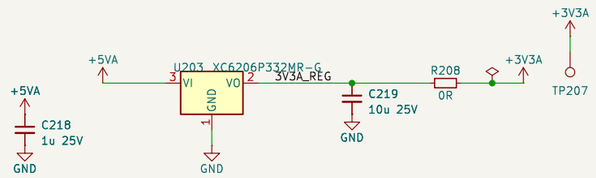
Vout = 0.596 V × (1 + 15k/2k) = 5.07 V; L and Cout from TI's table for 5 V. Input: own ferrite and 2 × 10 µF, keeping the 350 kHz input ripple off the ±12 V rail the op-amps share. EN divider 510k/75k: starts at ≈9.2 V, stops at ≈8.1 V, so the digital 5 V never runs while the analogue 5 V (L78M05) is down: TI's power-up order for the codec.



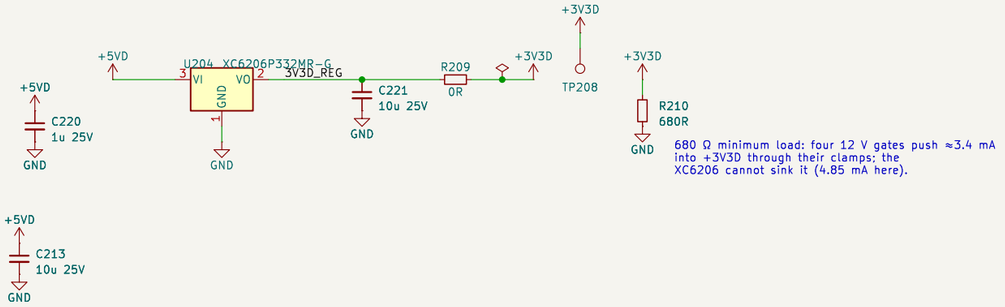
+5VA: L78M05 from +12 V, its own regulator (PCM3060 VCC, ADS8688 AVDD, +3V3A LDO)



+3V3A: XC6206 from +5VA (DAC8568, MCP3208, pot reference via POT_REF)

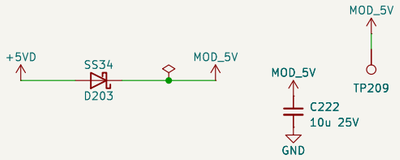


+3V3D: XC6206 from +5VD (codec VDD, ADS8688 DVDD, logic, Qwiic, MIDI)



MOD_5V: +5VD through a Schottky to Teensy VIN

Cut the Teensy's VIN–VUSB pads, so USB never back-feeds the buck and the buck never reaches USB.



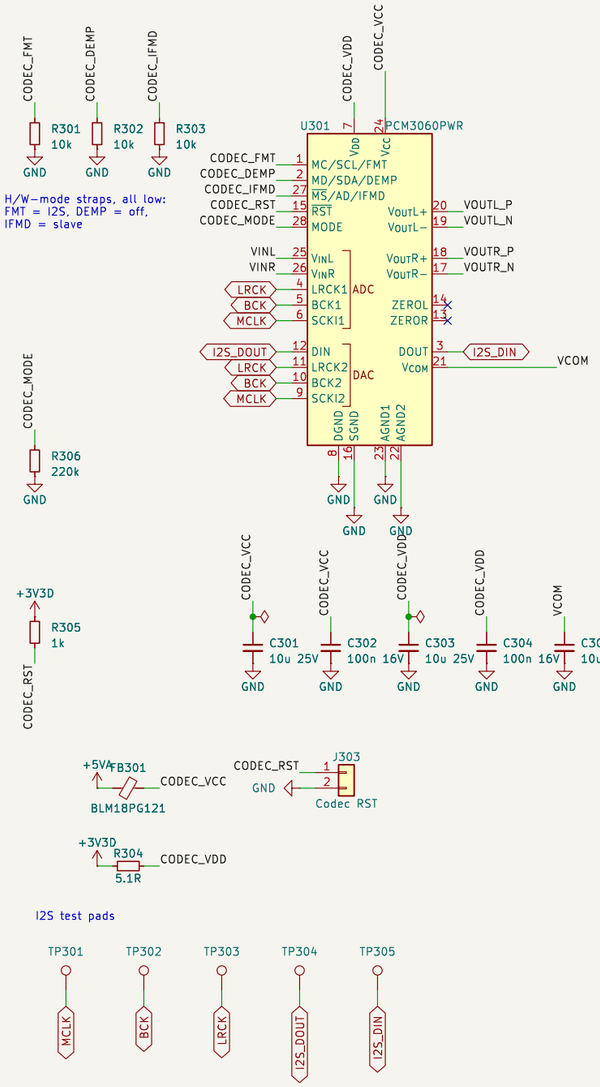
Load estimate: +12 V ≈ 120 mA typical; ~350 mA worst case with the Teensy and a lit LED chain.
−12 V ≈ 25 mA. XC6206: 200 mA each; +3V3D ≈ 50 mA.

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Analogue board for Teensy 4.1. Spec: docs/analogue_board_spec.md
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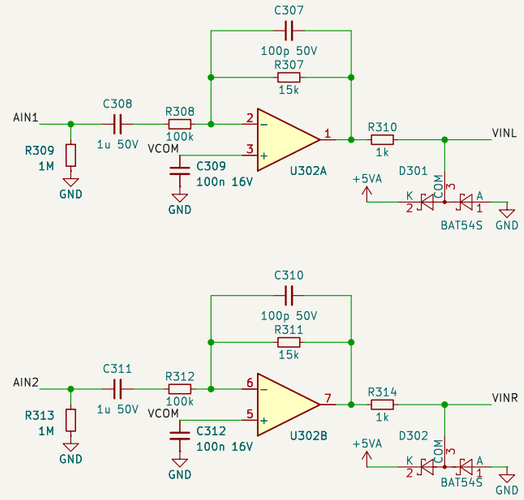
Sheet: /Power/		
File: power.kicad_sch		
Title: Power		
Size: A3	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 2/10

CODEC AND AUDIO

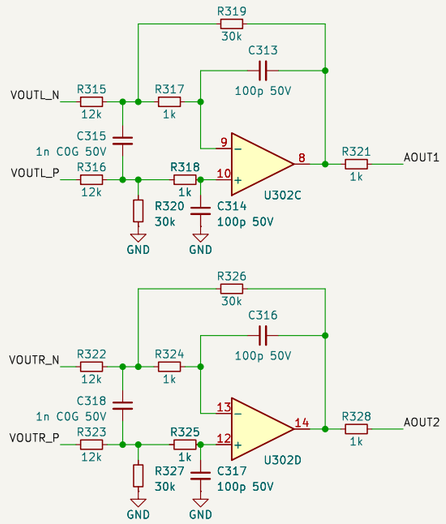
PCM3060 in hardware-control mode, the same straps as Mutable Beads: MODE 220 kΩ to DGND = parallel control with differential outputs; FMT low = 24-bit I2S, IFMD low = ADC and DAC both slaves, DEMP low = no de-emphasis. One I2S bus feeds both halves. RST is pulled up, so the codec starts by itself; CODEC_RST also reaches a hole (J303) so firmware can hold it in reset, as Beads does. Full scale is 20 Vpp in and out; firmware runs about 6 dB below. VCC via a ferrite from +5VA, VDD via 5.1 Ω from +3V3D: each with its own 10 µF + 100 nF.

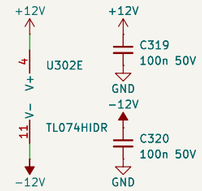


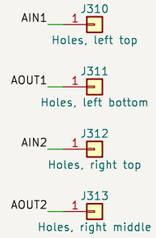
AUDIO INPUTS: 22 Vpp → 3 Vpp (codec full scale). AC-coupled, inverting (firmware flips the sign). ±12 V stays inside the codec rails, so the BAT54S clamps only act on faults. 100 nF on each + input decouples VCOM (12.5 kΩ source) at the op-amp, as the Patch SM does.



AUDIO OUTPUTS: balanced second-order MFB low-pass (Beads' topology), gain 30k/12k = 2.5, so 8 Vpp → 20 Vpp, DC-coupled, non-inverting (VOUT+ to the + side). 1 nF C0G across the two nodes and 100 pF from each op-amp input: f0 ≈ 65 kHz, Q ≈ 0.73, flat to 20 kHz (Basic parts except the 1 nF C0G). 1 kΩ 1206 out survives output-to-output patching.







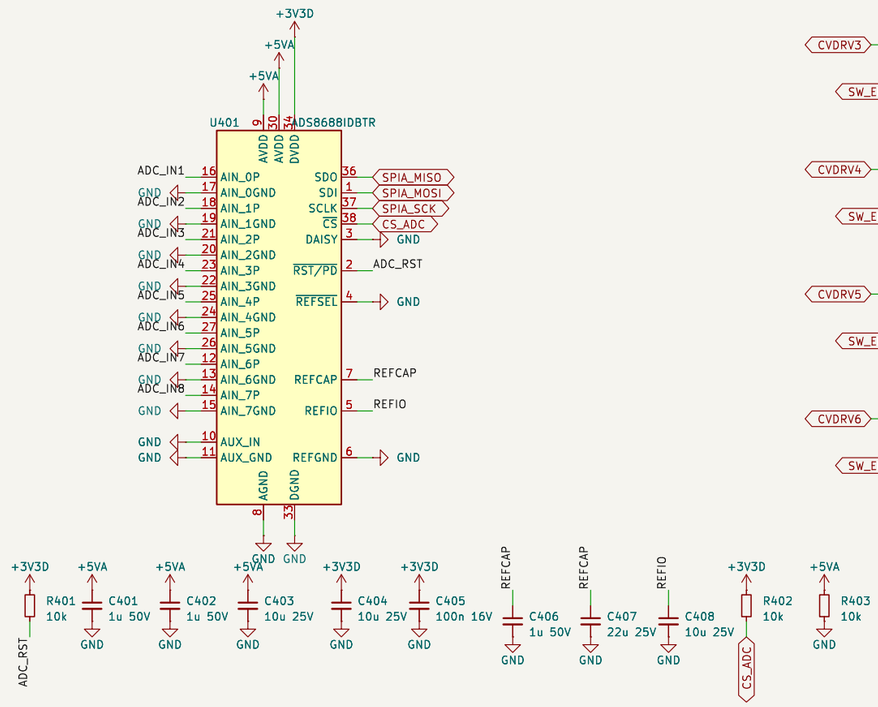
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Analogue board for Teensy 4.1. Spec: docs/analogue_board_spec.md

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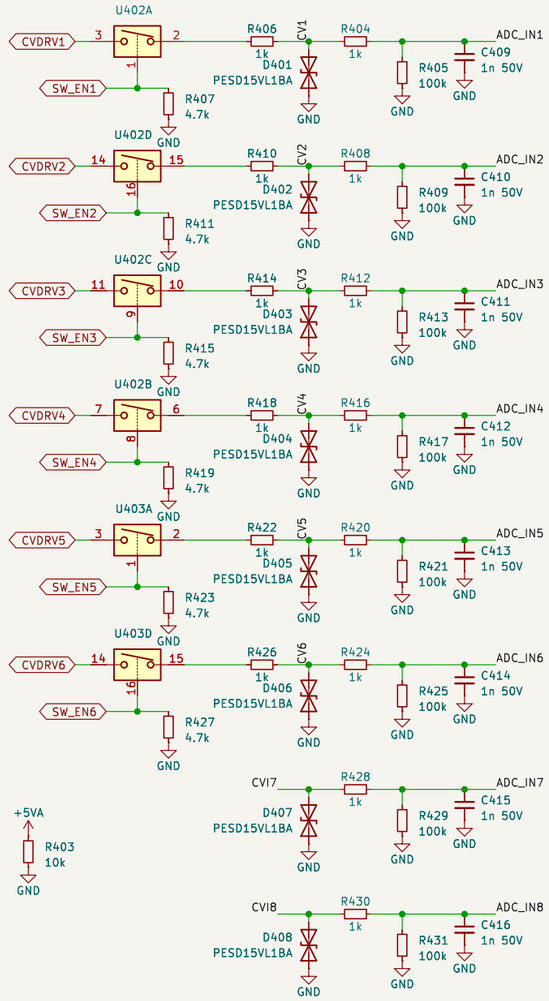
Sheet: /Codec & audio/		
File: codec_audio.kicad_sch		
Title: Codec & audio		
Size: A3	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 3/10

CV JACKS AND ADC

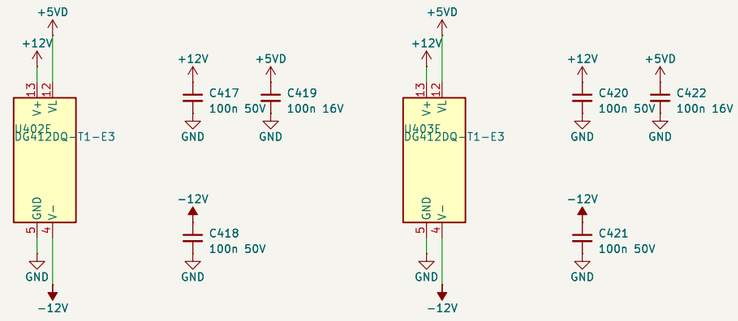
ADS8688: 8 channels, 16-bit, 1 MΩ inputs with ±10.24 / ±5.12 V ranges per channel (firmware default ±10.24 V), OVP ±20 V.
Each CV jack: 15 V TVS at the hole, read through 1 kΩ with 1 nF at the ADC pin (TI's EVM). 100 kΩ to ground, but the ADC's 1 MΩ returns to an internal bias, so an UNPATCHED input reads ≈ +0.2 V (no jack detect since 2026-10-16).
CV1-6 are bidirectional: a DG412 switch (on at logic 1) joins the DAC stage through 1 kΩ (1206); with it on the ADC reads the jack itself. 4.7 kΩ pull-downs beat the TCA9555's internal pull-ups: every jack starts as an input.



DAISY, REFSEL, AUX low: stand-alone, internal reference, AUX channel unused.
REFCAP 1 µF + 22 µF and REFIO 10 µF as TI's layout example.



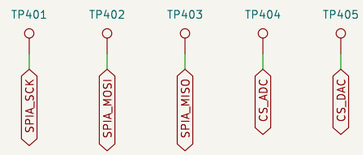
CVO7, CVO8 (out only): their 1 kΩ sits inside the DAC stage's feedback loop on the CV DAC sheet, so the jack holds its voltage under load. Calibrate them with a cable to CVI7 / CVI8.

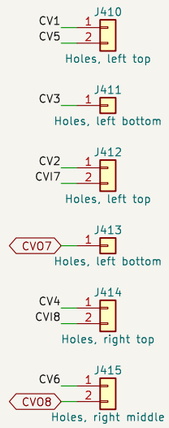


Spare switches (U403 SW3, SW4): off, both sides grounded



Converter bus test pads





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Analogue board for Teensy 4.1. Spec: docs/analogue_board_spec.md

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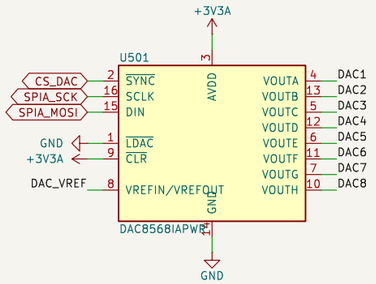
Sheet: /CV jacks & ADC/
File: cv_io.kicad_sch

Title: CV jacks & ADC

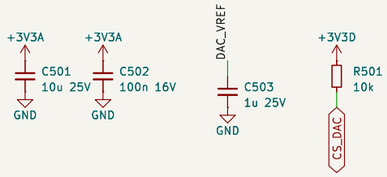
Size: A3	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 4/10

CV DAC

DAC8568 grade A: 0–2.5 V from its internal reference (off at power-up), outputs reset to zero. Each stage: non-inverting gain 1 + 33k/10k = 4.3, offset by OFFSET_REF = VREF × 22k/34k through the 10 kΩ: Vout = 4.3·Vdac − 5.34 V, so −5.34 V … +5.41 V, putting ±5 V inside the DAC's specified codes (0 V at code ≈ 32544). The offset tracks the DAC's own reference, as in Performer, O&C, Marbles and Stages. CVO7/8: the 1 kΩ sits inside the loop (feedback from the jack side, as Mutable does), so the jack holds its voltage under load. CV1–6 go through the DG412 switch; firmware corrects their load sag using the ADC, which reads the jack. ±10 V variant (not Rev A): Rf 75k, divider 15k/20k, all Basic: gain 8.5, OFFSET_REF 1.429 V, −10.71 … +10.54 V; needs ≥ 10 kΩ loads. The TL074H can swing it.

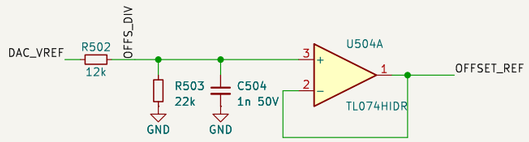


LDAC low: each write updates its output.
CLR high: clear pin unused.

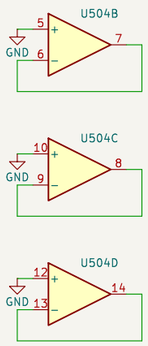


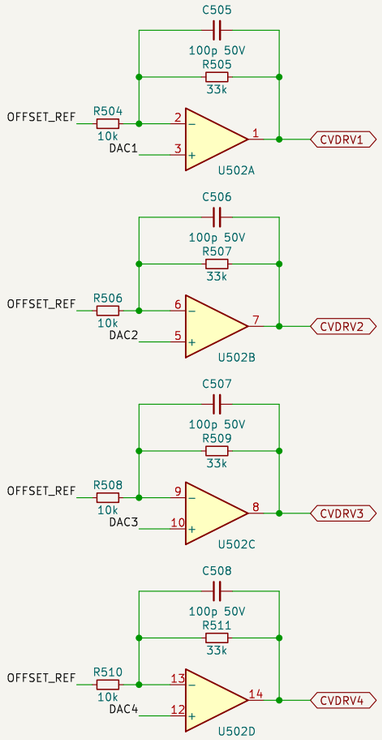
OFFSET_REF = VREF × 22k/(12k + 22k) = 1.618 V, buffered

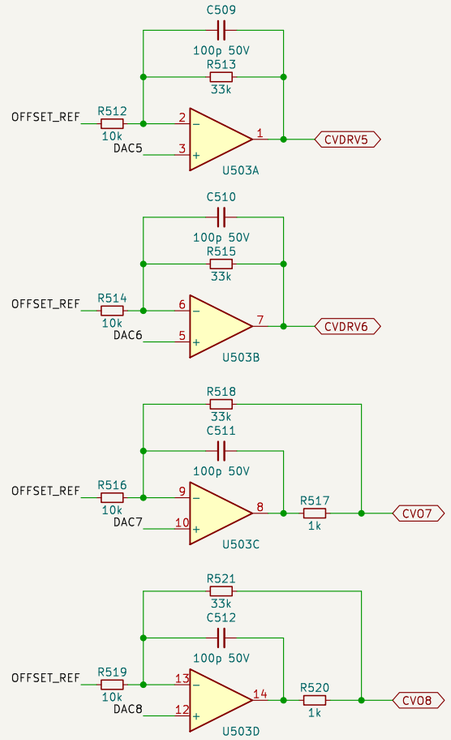
1 nF (not 1 µF): the offset must track VREF quickly, or enabling the reference glitches every output.



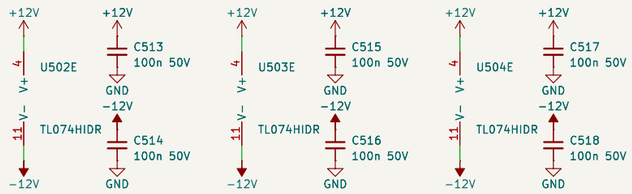
Spare U504 sections: followers with + to GND







CVDRV1–6 go to the DG412 switches (sheet CV jacks & ADC). CVO7–8 go straight to their holes.



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Analogue board for Teensy 4.1. Spec: docs/analogue_board_spec.md
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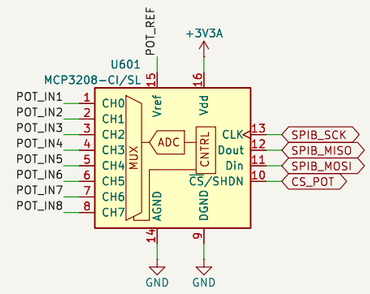
Sheet: /CV DAC/
File: cv_dac.kicad_sch

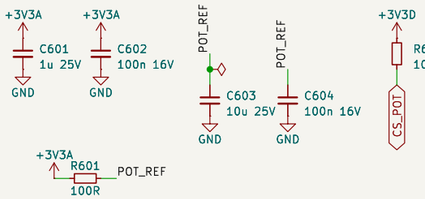
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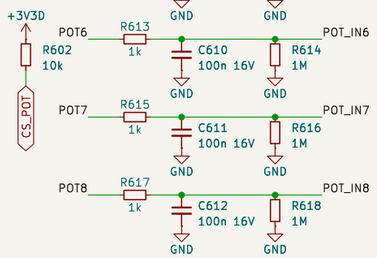
Size: A3	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 5/10

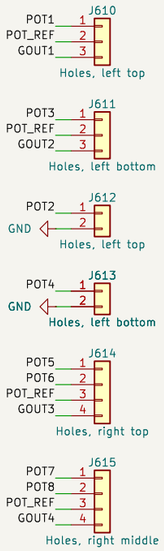
POTS AND GATE OUTPUTS

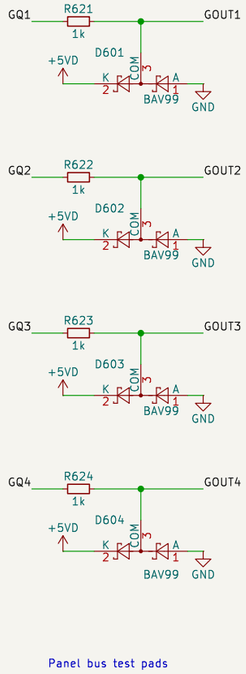
Pots: MCP3208 on +3V3A. Its VREF and the pot tops share POT_REF (100 Ω + 10 µF from +3V3A), so readings are ratiometric and the hand-wired pot leads can't disturb the DAC's supply. Each wiper: 1 kΩ + 100 nF, 1 MΩ to ground so an unwired channel reads 0 (100 kΩ would bend a 100 kΩ pot's response by up to 12 %).
Gate outs: four, one per wiring bank (2026-10-16), from QA–QD of a 74HCT595 at 5 V (VIH 2.0 V, so 3.3 V logic drives it) → 1 kΩ → jack, BAV99 clamps to +5VD and GND. QE–QH are unused; QH' still chains a second 595 from J603.
OE is held high (outputs off) by a pull-up until firmware drives GATE_OE_N low: the 595's register powers up random.



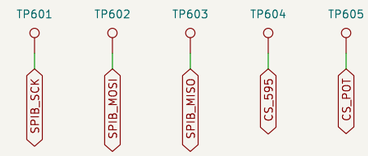


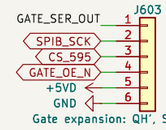






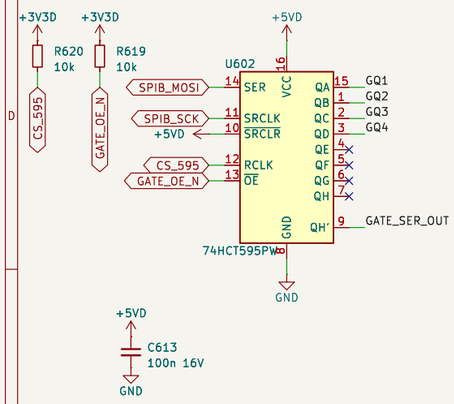
Panel bus test pads





Gate expansion: QH', SRCLK, RCLK, OE

Chain a second 74HCT595 off-board from J603 (74HCT595 only: a 74HC595 at 5 V needs 3.15 V inputs). OE keeps it off at boot too.



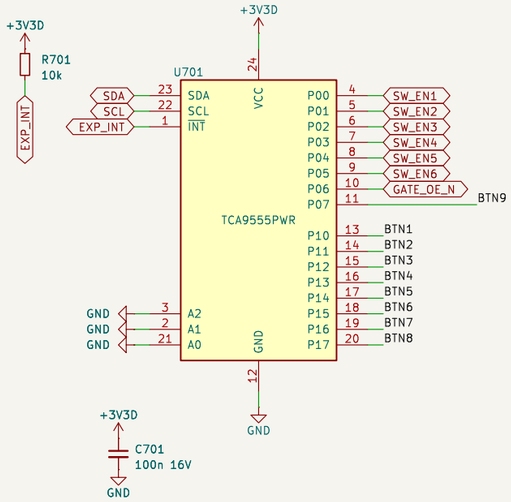
SRCLR tied high. Boot order: OE off, clear the register, then enable (firmware).

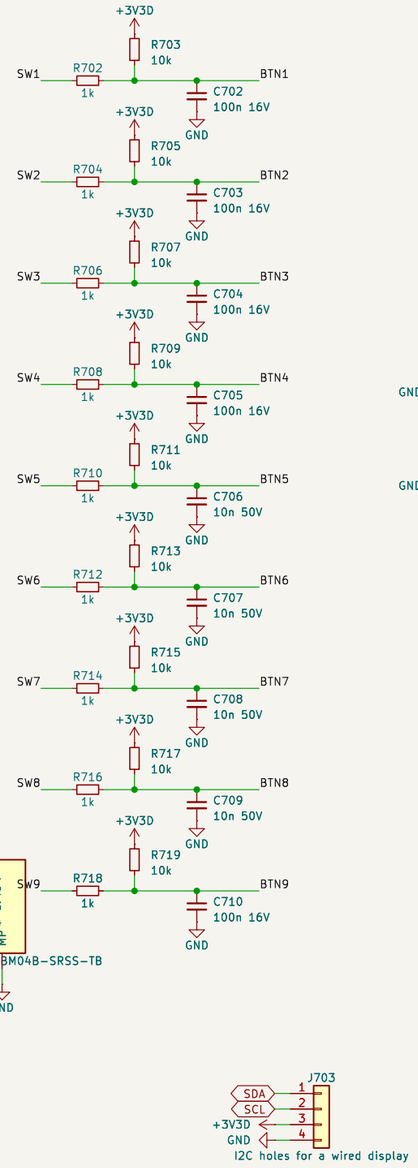
Generated by tools/analogue_board/build.py teensy: edit the generator, not these files
Analogue board for Teensy 4.1. Spec: docs/analogue_board_spec.md
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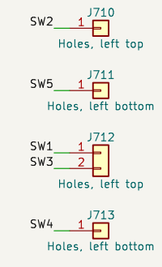
Sheet: /Pots & gate outs/		
File: pots_gates.kicad_sch		
Title: Pots & gate outs		
Size: A3	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 6/10

EXPANDER, I2C AND LEDS

TCA9555 at 0x20, M24C64 at 0x50; a Qwiic display is typically 0x3C/0x3D. 2.2 kΩ I2C pull-ups for 400 kHz.
The TCA9555 powers up with every line an input pulled high internally (~100 kΩ): switch enables and the gate
OE are defined by the pulls on their own sheets. Buttons and switches use P07 and P10–P17: hole → 1 kΩ → node with
10 kΩ to +3V3D and 100 nF (10 nF on SW5–SW8) → expander pin, as EuroPi does. A button pulls its hole to GND.
ENCODERS: wire A and B to gate inputs 3 and 4 (a clean comparator into a direct GPIO; encoder common to a +3V3D hole),
not to the expander: a busy OLED holds the shared I2C bus for ~23 ms and steps get lost. The push switch can use any SW hole.

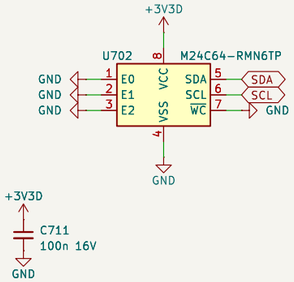




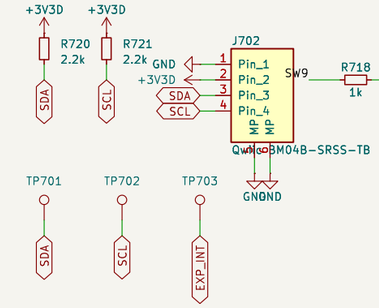


Encoder A/B: use gate inputs
3–4 (see note above); push
switch on any SW hole.

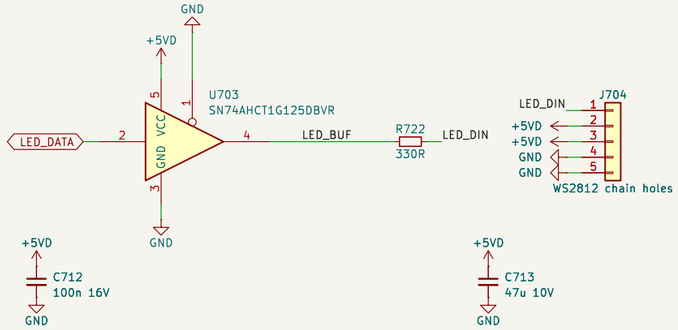
E0–E2 low: 0x50. WC low: writable.



Qwiic: 1 GND, 2 3.3 V, 3 SDA, 4 SCL



WS2812 data: 3.3 V → 74AHCT1G125 at 5 V → 330 Ω → LED_DIN. LED power from the +5VD holes.



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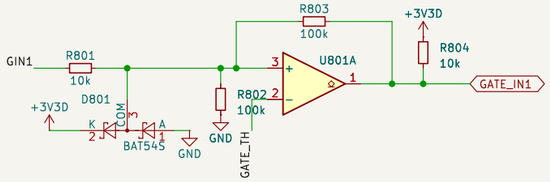
Sheet: /Expander, I2C & LEDs/
File: expander_i2c.kicad_sch

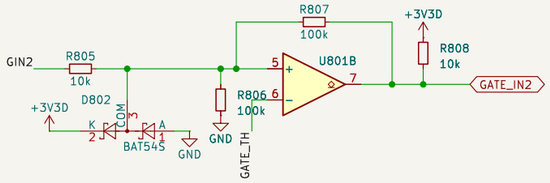
Title: Expander, I2C & LEDs

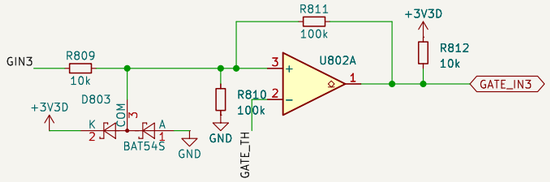
Size: A3	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 7/10

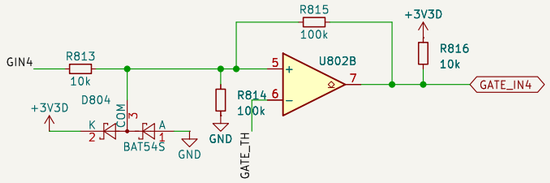
GATE INPUTS

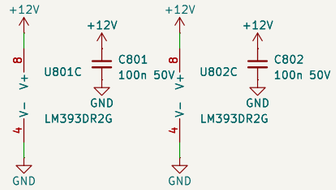
Adapted from Ogham's clock input (Ogham feeds the comparator's − input; this one is non-inverting, so four channels share one threshold divider): 10 kΩ, BAT54S clamps to +3V3D / GND, 100 kΩ to ground, LM393 against GATE_TH = 1.65 V with 100 kΩ hysteresis. Trips at ≈1.98 V rising, ≈1.67 V falling at the jack. The LM393 runs from +12 V so the clamped input (up to ≈3.6 V) stays inside its common-mode range (VCC − 1.5 V; at 5 V that would be 3.5 V). Open-collector output pulled up to +3V3D: GATE_INn is high while the gate is high, safe for any module. Firmware: catch 100 µs triggers with edge interrupts or PIO. The clamps push up to ≈3.4 mA into +3V3D: the 680 Ω minimum load on the power sheet absorbs it.



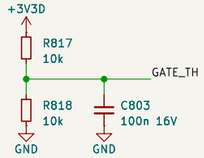


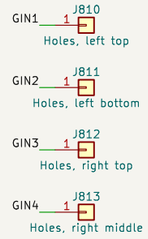






GATE_TH = +3V3D / 2





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Sheet: /Gate ins/		
File: gate_in.kicad_sch		
Title: Gate ins		
Size: A3	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 8/10

EDGE STRIPS

Two n8synth-style edge strips, 40 positions each at 2.54 mm. Each position is an outer hole (the deck's header pin, Dn) joined to an inner hole (a wiring point, Wn) by the net DECK_Ln / DECK_Rn: one pin per hole. Nothing else on the board connects to them: each module is wired by hand from the channel holes to the inner holes. Deck cells occupy positions 1–10, 16–25 and 31–40. The right strip serves only the 10HP deck: it has no holes at 11–15 and 26–30, and only the deck hole at 10, 16 and 31.
10HP: the board sits parallel behind the deck and both strips are used. 6HP: the board sits at 90° and only the left strip meets the deck's right-angle header.

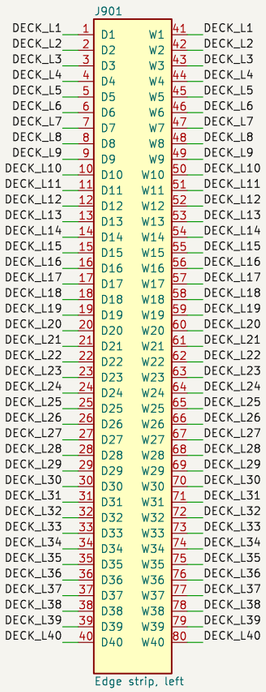
Fiducials for JLCPCB assembly

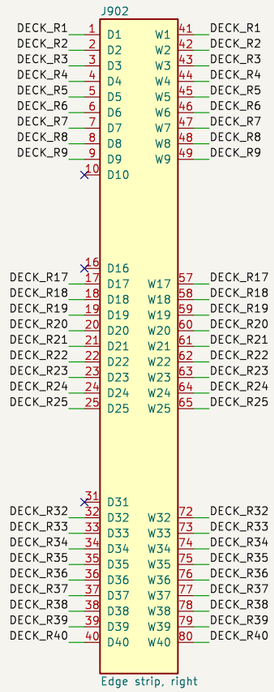
FID901 Fiducial

FID902 Fiducial

FID903 Fiducial

Hardware licence: CC BY-SA 4.0 (LICENSE in hardware/analogue_board).





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Sheet: /Edge strips/
File: edge_strips.kicad_sch

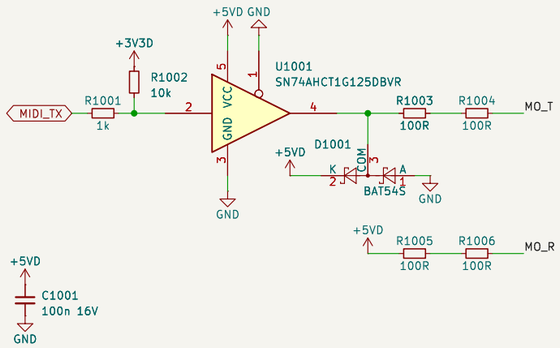
Title: Edge strips

Size: A3	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 9/10

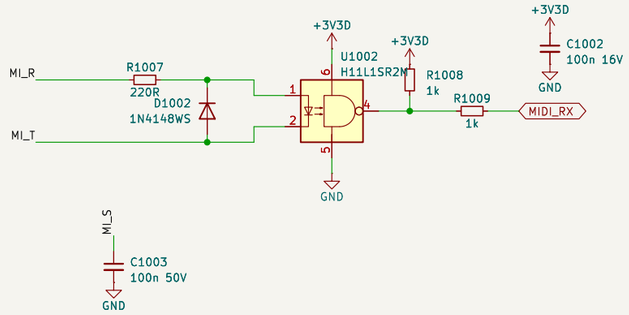
TRS MIDI IN AND OUT

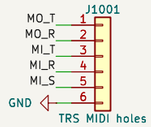
MIDI 1.0 over 3.5 mm TRS, Type A (MIDI Association RP-054): tip = current sink (DIN pin 5), ring = current source (DIN pin 4), sleeve = shield (DIN pin 2). Electrical values per CA-033 (2014). Uses the two UART-capable spare pins through link holes on J101 (SPARE1-MIDI_TX, SPARE2-MIDI_RX): left open, SPARE1/2 stay plain GPIO. Review: docs/analogue_board_review/midi.md

MIDI OUT: 3.3 V UART → 74AHCT1G125 at +5VD → 2 × 100 Ω per leg (1206) → TRS. Idle high = no loop current.
A mono cable shorting ring to sleeve draws 25 mA from the buck; each 100 Ω sees 64 mW.



MIDI IN: ring → 220 Ω → H11L1 LED → tip; 1N4148WS across the LED for reverse polarity. H11L1 switches at 1.6 mA (a worst-case 3.3 V sender still gives ≈4.5 mA). Output pulled up by 1 kΩ to +3V3D, then 1 kΩ to the link hole.
The IN jack's sleeve is NOT grounded (CA-033: no ground loop): wire it to MI_S (100 nF to ground) or leave it open.





WIRING (two stereo jacks in deck cells):
OUT: tip → MO_T, ring → MO_R, sleeve → the deck's D (ground) pad.
IN: tip → MI_T, ring → MI_R, sleeve → MI_S or open: NOT to D.
Turn MIDI on: link SPARE1-MIDI_TX and SPARE2-MIDI_RX on J101.
Type B gear: swap that jack's tip and ring wires.
Bring-up: loop OUT to IN with a stereo cable: ≈1.2–1.3 V across the 220 Ω while sending = 5.5–5.9 mA in the loop.
The H11L1 is ≈4 mm tall: place it outside the module footprint.

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Sheet: /TRS MIDI/
File: midi.kicad_sch

Title: TRS MIDI

Size: A3	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 10/10